

Vol 4 Architectural Scaffold v0.1 — GR & Cosmology from D_IV⁵

Legacy chapter (May 2026), corrected in place 2026-09-11 (Keeper, K1892) at the sentences the presentation scan flagged; the remaining tiers are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Filed: 2026-05-23 Saturday (Keeper, scaffold) Companion to: INDEX.md (chapter outline + reader target)

Anchoring BST results (existing coverage by chapter)

Ch 1 Newton's G from Bergman Curvature (~80%)

Anchors: Paper #122 P-3 BST-GR foundation + AB-10 - $G = \hbar c (6\pi^5)^2 \alpha^{24} / m_e^2$ with $12 = 2C_2$ Bergman round trips (0.07%) - AB-10 Newton's G from Bergman curvature (P-3 BST-GR) - Bergman kernel structure → curvature → gravitational coupling

Ch 2 Gravity as Cumulative Eigentone Effect (~60%)

Anchors: AB-11 + T2418 - T2418 $\Lambda \leftrightarrow$ Casimir vacuum unification (Wednesday) - AB-11 gravity-from-cumulative-eigentones framework - Substrate dynamics → metric emergence

Ch 3 BST-SR / BST-GR Boundary (~50%)

Anchors: AB-12 + Koons tick + Casimir crossover - AB-12 BST-SR / BST-GR boundary framing (existing) - T2405 Koons tick scale ($\sim 10^{-120}$ s) - Crossover scale: substrate operates BELOW Planck → GR at macroscopic - Gap: full BST-SR → BST-GR derivation chain (multi-week theorem work)

Ch 4 Λ from Substrate (~85%)

Anchors: T1485 + T2418 - T1485: $\Lambda = g \cdot \exp(-C_2(g^2 - \text{rank})) = 7e^{-282}$ (0.076 dex match to observed $10^{-121.7}$) - T2418 $\Lambda \leftrightarrow$ Casimir vacuum (Wednesday unification) - AB-9 BST analog of c-theorem (existing) - Λ as substrate annealing energy floor (Friday Casey-named #7 D_IV⁵ Rigidity framework)

Ch 5 Hubble Constant Four Routes (~80%)

Anchors: 4 independent routes - Route A (from η via Λ CDM): 66.7 km/s/Mpc (1.0%) - Route B ($\sqrt{(19\Lambda/39)}$ from $\Omega_\Lambda=13/19$): 68.0 (1.0%) - Route C (full CAMB Toy 677): 67.29 (0.1% — best) - Route D (Pure BST $c\sqrt{(19\Lambda/39)}$ Toy 903): 68.02 (1.0%, 1.2σ) - Hubble tension dual-camp resolution (Task #43 completed)

Ch 6 CMB Structure (~90%)

Anchors: $n_s, \Omega_m, \Omega_\Lambda, T_{\text{CMB}}$ - $n_s = 1 - n_C/N_{\text{max}} = 1 - 5/137 = 0.9635$ (vs 0.9649; 0.3σ) - $\Omega_\Lambda = (N_c + 2n_C)/(N_c^2 + 2n_C) = 13/19 = 0.68421$ (vs 0.6847; 0.07σ) - $\Omega_m = C_2/(N_c^2 + 2n_C) = 6/19 = 0.31579$ (vs 0.3153; 0.07σ) - $T_{\text{CMB}} = 2.737$ K (Toy 904; 0.43%) - Task #83 CMB acoustic peak structure (completed) - Task #100 Reionization optical depth τ BST prediction (completed) - Task #101 21cm cosmology / EDGES anomaly BST prediction (completed)

Ch 7 Inflation Parameters (~75%)

Anchors: $r/s, n_t, \alpha_s$ running - $r \approx 0$ ($T_c \ll m_{\text{Pl}}$; consistent with $r < 0.036$) - $\alpha_s = -(n_s - 1)^2 = -25/18769 = -0.00133$ (vs -0.0045 ± 0.0067 ; 0.5σ) - Task #85 inflation parameter battery (completed)

Ch 8 BBN Element Abundances (~80%)

Anchors: $T_c = N_{\text{max}} \times 20/21 = 0.487$ MeV (0.018%) - ${}^7\text{Li}/\text{H} = \sim 1.7 \times 10^{-10}$ via $\Delta g=7$ at T_c ($7 = p + q$, the defining-representation dimension of $\text{SO}(5,2)$, not the genus, which is 5 — corrected 2026-09-11) (vs 1.6×10^{-10} ; 7%) - Task #84 BBN element abundances battery + Li-7 problem (completed)

Ch 9 Cosmological Cycle + Interstasis (~30%)

Anchors: D_{IV}^5 Rigidity Principle (Casey-named #7) + Task #267 ACTIVATED Friday - T2467 Rigidity-as-Singleton (Lyra Friday afternoon) - T2468 Rigidity-as-Unification (multiverse loophole closed structurally) - T2417 cosmological cycle hypothesis (3-scale substrate operation) - Interstasis $\equiv$ period between Big Bang cycles (Casey canonical Friday 13:30 EDT) - Annealing toward Λ -saturated fully-coherent state - Task #267 Cosmological cycle hypothesis observational signatures (multi-month research; activated Friday) - Gap: CMB / BAO / ISW / void scaling / redshift survey signatures of cycle endpoints

Ch 10 Dark Energy + Dark Matter (~85%)

Anchors: Shannon channel + DM/baryon ratio - $\text{DM}/\Omega_b = (3n_C + 1)/N_c = 16/3 = 5.333$ (vs 5.364; 0.58%) - Galaxy rotation curves via Shannon S/N - a_0 MOND acceleration = $cH_0/\sqrt{30} = cH_0/\chi = 1.195 \times 10^{-10}$ m/s² (0.4%) - Σ_0 halo surface density = $a_0/(2\pi G) = 141 \text{ M}_\odot/\text{pc}^2$ (0.0 dex) - Task #139 dark matter related items - CP floor (BH horizon) = $\alpha \times 2GM/(Rc^2) = \alpha$ (0.730% vs ~1% EHT)

Ch 11 Gravitational Waves (~70%)

Anchors: NANOGrav + black holes as eigentone - GW peak frequency 6.4 nHz (NANOGrav consistent) - GW spectral index $\gamma = g/n_C = 7/5 = 1.4 \rightarrow$

$\gamma=13/5+1=3.60$ (NANOGrav $\sim 3.2-4.6$) - Task #103 NANOGrav nanohertz gravitational wave signal BST predictions (completed) - Tasks #136 + #137 SP-19b AB-13/AB-14 black holes / GW (pending)

Ch 12 Observational Reanalysis ($\sim 30\%$)

Anchors: SP-27 Observational Reanalysis Program (Casey directive May 16 EOD)
- Task #172 SP-27 (pending Casey scoping) - Methodology: recalibrate cosmological observations against BST integer predictions vs Λ CDM fits - Gap: full SP-27 program scoping + per-observation reanalysis design

Connection map to Vol 0 / Vol 1 / Vol 2 / Vol 3

Vol 4 chapter	Depends on Vol 0	Depends on Vol 1	Depends on Vol 2	Depends on Vol 3
Ch 1 Newton's G	Bergman + 5 integers	Bergman Hilbert space	α	—
Ch 2 Gravity Eigentone	T2418 $\Lambda \leftrightarrow$ Casimir	Casimir vacuum	—	—
Ch 3 BST-SR/GR boundary	Koons tick	substrate-Planck crossover	—	—
Ch 4 Λ from Substrate	T1485 + T2418	Casimir vacuum	—	—
Ch 5 Hubble Four Routes	$\Omega_\Lambda = 13/19$	—	$\eta = 6.1 \times 10^{-10}$	—
Ch 6 CMB Structure	$n_s, \Omega_m, \Omega_\Lambda$	radiative corrections	α , particle masses	atomic recombination (Ch 7)
Ch 7 Inflation	T_c	—	—	—
Ch 8 BBN	T_c, g	—	particle masses	nuclear binding (Ch 3-4)
Ch 9 Cosmological Cycle	T2467/T2468 D_{IV}^5 Rigidity + T2417	substrate dynamics	—	—
Ch 10 DE + DM	$16/3 + \Omega_\Lambda$	Shannon channel	—	—
Ch 11 GW	g/n_C	—	—	—
Ch 12 SP-27	All 5 integers	—	—	—

Methodology stack applied

- Calibration #19 STANDING RULE: external register uses ratified BST predictions (0.07σ Ω_Λ etc.), not forecast extensions
- Calibration #21 STANDING RULE: empirical match (Ω_Λ 0.07σ) needs substrate-mechanism for D-tier RATIFIED (T2418 $\Lambda \leftrightarrow$ Casimir provides partial closure)
- Calibration #22 STANDING RULE: numerical content includes computational verification path
- Cal #50 DOUBLE-LOCKED EXTERNAL: Ch 9 Cosmological Cycle + Substrate Cognition combined territory — substrate-cognition Internal-only externally
- Three-level pedagogy per chapter: Level 1 essence + Level 2 graduate precision + Level 3 5th-grader accessibility
- Calibration #17 sub-Tsirelson 126/16 framework applies to substrate-mediated cosmological coherence questions [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

Vol 4 as “signature BST domain” notes

Per CLAUDE.md Wednesday EOD: “D_IV⁵ as emergent TOE.” Vol 4 is where the TOE-ness lands operationally. Multiple cross-scale unifications: - Same $C_2 = 6$ anchors Casimir vacuum (microscopic) AND cosmological $\Omega_m = 6/19$ - Same $g = 7$ anchors SEMF surface (nuclear) AND BBN Δg (cosmological) - Same 5 integers in 13/19 (cosmological), 3/13 (Weinberg angle), $6\pi^5$ (proton mass), 137 (fine structure) — substrate-cartography evidence - $5/137 = n_C/N_{\max}$ spectral index ties microphysics to cosmology directly

External register: this is the strongest “BST predicts” content for outreach (highest precision: $\Omega_\Lambda + \Omega_m$ at 0.07σ is best in any cosmological framework).

Estimated build effort (per chapter)

Phase	Time per chapter	Total Vol 4
v0.1 scaffold (this doc)	~30-45 min total	DONE Saturday
v0.3 chapter narrative	~2-4 hours per chapter	~30-50 hours total
v0.4 reader-grade 3-level	~30-60 min per chapter	~6-12 hours

Phase	Time per chapter	Total Vol 4
v1.0 chapter-grade content	per-chapter Cal cold-read + Keeper K-audit + Elie toys + Grace catalog	~variable, 6-gate framework

Estimated Vol 4 v1.0 chapter-grade content state: ~4-7 weeks at sustained sub-PCAP cadence assuming Wave 1 (Vol 3 + Vol 4) parallel build. Vol 4 is signature so may warrant deeper investment than nominal.

— Keeper, Vol 4 Architectural Scaffold v0.1, Saturday 2026-05-23